Fracture graphs in the strict high-rank range

Human Compression project: a component of the L019 recovery

Write $G_i = \langle \rho_j : j \neq i \rangle$ for a maximal omission of a string C-marking $\rho_0, \dots, \rho_{r-1}$. A fracture graph chooses, for every i, one transposition of ρ_i whose endpoints belong to different G_i-orbits. We recover the initial fracture argument needed for the stability theorem, with the degree-six action qualification made explicit.

Lemma 1. *Let $m \geq 5$ be odd. A transitive string C-marking generating the natural S_{2m} cannot have a maximal omission of rank $m+1$ with the exceptional $C_2 \wr S_m$ marking of the complete transitive half-degree classification.*

Proof. Let the omission be $H = G_i$, preserving its m pairs. The acquired classification gives its actual base marks $\lambda, v = \lambda + e_1$, where λ flips all pairs. Its other $m-1$ marks $s_1, \dots, s_{m-1}$ form a complement with block images the adjacent transpositions of a path. The lifts can contain a cocycle; we do not replace the action by its abstract cocycle normalization. In the actual action, conjugation on the base still permutes coordinates as the block image does. The noncommutation diagram of H is the isolated mark λ and the path

$$v, s_1, \dots, s_{m-1}. \tag{1}$$

The diagram of the parent S_{2m} is connected. Indeed a disconnection, together with IP, would express this group as a direct product of two nontrivial commuting normal subgroups. This is impossible for S_d, $d \geq 5$: every nontrivial normal subgroup contains A_d, or else centralizes A_d and is trivial; the natural centralizer of A_d is trivial. Thus every adjacent pair in the parent string fails to commute. Deleting position i leaves paths of lengths i and $r-i-1$. To obtain the two components above, the omitted mark $q = \rho_i$ must lie between λ and an endpoint of (1). In particular $i = 1$ or $r-2$, rather than an endpoint of the whole string.

If the adjacent endpoint is v, then q centralizes $B = \langle s_1, \dots, s_{m-1} \rangle$. Apply the acquired actual pair-centralizer lemma: a group transitive on the pairs, with primitive block image generated by involutions, has its centralizer inside the pair stabilizer, except for a regular prime-dihedral action. Here the block image is S_m, so that exception is impossible. Consequently q preserves the pairs.

If the adjacent endpoint is s_{m-1}, then q centralizes $v, s_1, \dots, s_{m-2}$. Conjugating v by the latter block path gives all the actual base vectors

$$\lambda + e_j \quad (1 \leq j < m).$$

Their subgroup N has exactly the original pairs as its point orbits. Every coordinate occurs nontrivially: coordinate m occurs in all these vectors, and each coordinate $j < m$ occurs in one with a different index. Since q centralizes N, it permutes its orbits and again preserves the pairs. In both cases $\langle H, q \rangle$ preserves a nontrivial partition, contradicting its being the natural S_{2m}. □

Lemma 2. *Every connected rank-six C-marking of the natural S_8 has all its maximal omissions intransitive. For every rank-six C-marking of the natural A_{11}, the centralizer in S_{11} of either five-mark endpoint omission is trivial.*

Proof. These are finite consequences of two acquired complete classifications, not assumptions about generic rank bounds. The accepted degree-eight certificate has exactly two ordered conjugacy classes, the following marking and its reversal:

$$(2\ 3),\ (1\ 2)(3\ 4),\ (4\ 5),\ (5\ 6),\ (6\ 7),\ (7\ 8). \tag{2}$$

Every omission in (2) is intransitive. Omitting its first mark separates $\{1,2\}$ from the other points; omitting its second fixes 1; omitting a later mark separates the corresponding terminal segment of the displayed path. This proves the first assertion, including the reverse order.

The complete acquired A_{11} census has four rank-six ordered representatives. They have common entries

$$\begin{aligned} t_0 &= (8\ 9)(10\ 11), \quad t_1 = (7\ 8)(9\ 10), \\ t_3 &= (3\ 4)(5\ 6), \qquad t_5 = (1\ 2)(3\ 4), \end{aligned}$$

with independently chosen entries

$$\begin{aligned} t_2 &= (6\ 7)(10\ 11) \text{ or } (6\ 7)(8\ 9), \\ t_4 &= (2\ 3)(4\ 5) \text{ or } (1\ 3)(2\ 5). \end{aligned} \tag{3}$$

For all eight endpoint omissions, the literal-permutation certificate finds exactly one commuting permutation, the identity. Its exhaustive enumeration assigns the image of an unassigned point, then propagates that choice under each specified mark. A conflict or a repeated image rejects the choice. Every unused image is tried at every new orbit, so every equivariant bijection is included. A final direct commutation test checks each completion. We reuse the accepted point-fixing implementation by adjoining one new fixed point and fixing it: restriction gives exactly the unrestricted centralizer on the original eleven points. A separate GAP centralizer computation confirms all eight orders. The same replay records the twelve omission orbit partitions for the two literal S_8 representatives. Complete input arrays, extraction pins and both executions are retained. □

Theorem 3. *Let $G \le S_n$ be transitive, with a string C-marking of rank r, and assume $n \ge 7$ and $2r > n+2$. Then $G = S_n$ and every G_i is intransitive. Every choice of one crossing transposition for each label is a fracture graph and is a forest. For $n = 5, 6$ the same conclusion holds after choosing the ordinary degree-n action in its marked isomorphism class.*

Proof. We acquire the complete transitive half-degree bound (L047) and the exact alternating maxima (L044). A transitive group different from A_n, S_n cannot have this rank: its exceptional wreath rank is $n/2+1$, and the exceptional small actions also lie below the strict threshold. The alternating maxima are $4, 5, 6$ for $n = 9, 10, 11$, no C-marking exists for $n = 7, 8$, and the maximum is $\lfloor (n-1)/2 \rfloor$ for $n \ge 12$. Thus $G = S_n$.

Suppose G_i is transitive. IP makes it proper in G and makes its inherited $r-1$ marks an independent C-marking. Since $2(r-1) > n$, the same two classifications leave only the following possibilities:

1. $n = 2m$, with $m \ge 5$ odd, and $G_i = C_2 \wr S_m$ has rank $m+1$;
2. $n = 8$, $r = 6$, and G_i is the degree-eight imprimitive exception of rank five;
3. $n = 11$, $r = 7$, and $G_i = A_{11}$ has rank six.

Lemma 1 excludes the first case. In the second, the parent diagram is connected, so the complete parent classification in Lemma 2 applies and contradicts transitivity of G_i. In the third, the omission diagram is connected because A_{11} is simple. Since the parent diagram is a path, i is an endpoint. The omitted involution must centralize the five marks distant from it, contrary to Lemma 2. All omissions are therefore intransitive.

For each i, some transposition of ρ_i joins different G_i-orbits; otherwise every generator would preserve these orbits and G would be intransitive. Choose one. If the chosen edges had a cycle, deleting its i-edge would leave a path between that edge's endpoints using only labels different from i. Such a path lies in a single G_i-orbit, contradicting the choice. Thus the selected graph is a forest.

For $n = 5, 6$, the same complete rank bounds force $G = S_n$. The independent-family bound $r \leq n - 1$ forces equality. The acquired complete small extremal classification gives only the simplex marking up to marked-group isomorphism. Choose its ordinary adjacent-transposition action. Its omission orbits are the two intervals separated by the omitted transposition, and the path itself is a fracture graph. □

Action qualification and attribution. The fixed source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, Proposition 4.1. The rank classifications, the original stability claim and the fracture construction remain attributed to their authors. The recovered argument supplies the actual pair-centralizer justification, preserves cocycle twists in the permutation action, and states the small action qualification. In degree six the retained census also contains the outer image of the simplex: both its endpoint omissions are transitive. Thus a literal assertion about every degree-six action would be false. This does not change the source's intended counting by marked-group isomorphism, for which the ordinary simplex is a valid representative.

The pair-centralizer lemma is the generic first lemma of the accepted L044 component; its subsequent alternating-group theorem is not used here. The finite premises are the complete accepted degree-eight rank-six census, the complete A_{11} census, and the complete degree-five and six extremal census. Only the stated finite consequences are newly checked. This component proves neither perfect-split existence nor the stability bijection or its six constants. It grants no whole-original or open-conjecture credit and adds no Lean or compression measurement.